

SCHOOL MANAGEMENT SYSTEM FOR MATUNDU PRIMARY SCHOOL

WORK IN PROGRESS DOCUMENT

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Registration Number: 23/05037

Submission Date: August 2026

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A project document submitted in partial fulfillment of the requirements for Bachelor of
Information Technology (BIT 3105)

DECLARATION

I declare that this work in progress document is my original work and has not been submitted elsewhere for examination.

Signature: _____ Date: _____

Mwange Musa Muthami (23/05037)

Supervisor Approval:

Signature: _____ Date: _____

Dr. Simon N. Mwendia / Dr. Kevin Mugoye Sindu

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1. PROJECT OVERVIEW

1.1 Project Title

School Management System for Matundu Primary School

1.2 Student Details

Field	Detail
Student Name	Mwange Musa Muthami
Registration Number	23/05037
Course	Bachelor of Information Technology (BIT 3105)
Supervisors	Dr. Simon N. Mwendia / Dr. Kevin Mugoye Sindu
Academic Year	2025/2026
Institution	KCA University

1.3 Project Description

The Matundu Primary School Management System (MATUNDU-SMS) is a production-grade, full-stack web application designed to digitize and streamline all administrative and academic operations at Matundu Primary School in Voo, Kitui County. The system is fully aligned with Kenya's Competency-Based Curriculum (CBC) and handles the complete lifecycle of school operations including student admissions, attendance tracking, CBC-aligned assessments, report card generation, fee management, timetabling, and stakeholder communication.

1.4 Problem Being Solved

Matundu Primary School has historically relied on paper-based record keeping for all administrative functions. This creates critical challenges:

- Data loss and corruption of physical student records
- Manual computation of CBC competency grades consuming excessive teacher time

- No real-time access for parents to track student progress and fee balances
- Slow and error-prone fee reconciliation by the bursar
- Inability to generate standardized CBC-compliant report cards quickly

2. TECHNOLOGY STACK

Layer	Technology	Version
Frontend	React 18 + Vite + Tailwind CSS	18.3 / 7.3 / 3.4
State Management	Zustand	5.0.1
Charts	Recharts	2.13.3
HTTP Client	Axios	1.7.7
Router	React Router DOM	6.28.0
PDF (Frontend)	jsPDF + AutoTable	4.1.0 / 5.0.7
Backend	Node.js 20 + Express.js	20+ / 4.21.1
ORM	Prisma	5.22.0
Database	PostgreSQL	15+
Cache	Redis	5.10.0
Auth	JWT + bcrypt	—
PDF (Backend)	PDFKit	0.15.0
Logging	Winston	3.15.0
Security	Helmet	8.0.0
Containerization	Docker + Compose	—
Deployment (FE)	Cloudflare Pages	—
Deployment (BE)	Render	—

3. PROJECT ARCHITECTURE

3.1 System Architecture Overview

The system employs a three-tier client-server architecture deployed on modern cloud infrastructure:

Tier	Technology	Hosting	Purpose
Client (Frontend)	React 18 SPA	Cloudflare Pages	User interface for all roles
Server (Backend)	Express.js REST API	Render	Business logic and data access
Database	PostgreSQL 15	Render (managed)	Persistent relational data storage
Cache	Redis	Render (managed)	Rate limiting and session caching

3.2 Project Folder Structure

matundu-sms/

- └─ backend/src/app.js # Express entry point (32KB)
- └─ backend/src/config/ # DB, Redis, env config
- └─ backend/src/middleware/ # Auth, RBAC, audit, sanitize
- └─ backend/src/features/ # 13 feature modules
- └─ backend/prisma/schema.prisma # 24-model DB schema
- └─ frontend/src/App.jsx # React router root
- └─ frontend/src/pages/ # 17 page components
- └─ frontend/src/components/ # Reusable components

- |— frontend/src/stores/ # Zustand global state
- |— frontend/src/services/ # Axios API wrappers
- |— docker-compose.yml # Local dev environment
- └— render.yaml # Cloud deployment config

4. DATABASE DESIGN

4.1 Schema Overview

The Prisma schema defines 24 models and 6 enums covering all functional domains of the school management system.

4.2 Core Models

Model	Purpose	Key Fields
User	Base authentication entity	id, email, password, role, isActive
Student	Learner profiles	admissionNumber, firstName, classId, admissionStatus
Staff	Teacher/admin profiles	employeeNumber, firstName, qualification, specialization
Guardian	Parent/guardian details	firstName, lastName, relationship, nationalId
AcademicYear	School year definition	name, startDate, endDate, isCurrent
Term	Term within academic year	termNumber, startDate, endDate, academicYearId
Grade	CBC grades PP1-Grade 9	name, level
Stream	Class streams	name
Class	Grade+Stream+Year combo	name, gradeId, streamId, academicYearId
Subject	Subject definitions	name, code, gradeId
Attendance	Daily attendance records	date, status, studentId, classId, termId
Assessment	Test/exam definitions	name, type, maxScore, weight, subjectId, termId

AssessmentScore	Student scores	score, grade, studentId, assessmentId
Competency	CBC competencies	name, description, subjectId
CompetencyScore	CBC ratings per student	rating (EE/ME/AE/BE), studentId, competencyId
ReportCard	Termly report	totalScore, averageScore, rank, pdfUrl
FeeStructure	Fee config per grade/term	name, amount, gradeId, termId
StudentInvoice	Student billing	invoiceNo, totalAmount, paidAmount, balance
Payment	Payment records	amount, method, status, transactionRef, mpesaReceiptNo
Announcement	School announcements	title, content, targetRoles, isPublished
Message	Direct messages	subject, content, senderId, receiverId
TimetableSlot	Weekly timetable	dayOfWeek, startTime, endTime, classId, subjectId
AuditLog	System audit trail	action, entity, userId, ipAddress
AiChatMessage	AI assistant history	role, content, studentId

4.3 Enumerations

Enum	Values
Role	SUPER_ADMIN, ADMIN, TEACHER, STUDENT, PARENT, BURSAR
AdmissionStatus	PENDING, APPROVED, REJECTED
AttendanceStatus	PRESENT, ABSENT, LATE, EXCUSED

CompetencyRating	EXCEEDING (EE), MEETING (ME), APPROACHING (AE), BELOW (BE)
PaymentMethod	CASH, MPESA, BANK_TRANSFER
PaymentStatus	PENDING, COMPLETED, FAILED, REFUNDED

5. BACKEND IMPLEMENTATION

5.1 Middleware Pipeline

Order	Middleware	File	Purpose
1	CORS	cors	Cross-origin request handling
2	Helmet	helmet.js	Security headers (XSS, CSRF, clickjacking)
3	Rate Limiter	express-rate-limit + Redis	Brute-force protection
4	Request ID	requestId.js	Unique request tracking
5	Body Parser	express.json()	JSON request parsing
6	Sanitize	sanitize.js	Strip HTML/script injection
7	Auth (JWT)	auth.js	JWT token verification
8	RBAC	rbac.js	Role-based permission enforcement
9	Validate	validate.js	express-validator result handling
10	Audit Log	auditLog.js	Mutation activity recording
11	Error Handler	errorHandler.js	Global error formatting

5.2 Feature Modules Implemented

Module	Files	Key Functionality
auth	4 files	Login, register, JWT refresh, profile, password change
students	4 files	CRUD, admission workflow, guardian linking, class assignment
staff	3 files	Staff CRUD, teacher-class-subject assignments
academic	12 files	Years, terms, grades, classes, subjects, timetable, outlines, resources
assessments	3 files	Create assessments, bulk score entry, CBC grading
attendance	5 files	Daily attendance marking, QR code attendance service
reports	3 files	Report card generation, class rankings, PDF export
fees	4 files	Fee structures, invoice generation, payments, M-Pesa service
communication	3 files	Announcements with role targeting, direct messaging
ai	3 files	AI chat assistant service with conversation history
reminders	3 files	Student reminders with Google/Microsoft calendar sync
integrations	3 files	Google Calendar OAuth,

		Microsoft Tasks via Graph API
dashboard	—	Aggregated KPIs for admin and student dashboards

6. FRONTEND IMPLEMENTATION

6.1 Pages Implemented

Page	File Size	Role Access	Description
Login	3.9 KB	Public	JWT authentication form
Dashboard	40.2 KB	All Roles	Analytics hub with KPI cards and Recharts
Students	11.9 KB	Admin/Teacher	Student list with search and filter
StudentDetail	20.3 KB	Admin/Teacher	Full student profile, attendance, scores, fees
Staff	11.8 KB	Admin	Staff management and assignment
Classes	14.7 KB	Admin/Teacher	Class and academic structure management
Assessments	21.6 KB	Teacher	Score entry with auto-grading
Attendance	10.8 KB	Teacher	Daily attendance marking interface
Fees	21.3 KB	Bursar/Admin	Invoice management and payment recording
Reports	50.4 KB	Admin/Teacher	Report card generation with PDF export
Timetable	22.1 KB	Admin/Teacher	Visual weekly

			timetable grid
Announcements	12.9 KB	All	School-wide announcements
Admissions	10.2 KB	Admin	Admission approval/rejection workflow
Settings	22.7 KB	Admin	System configuration
Profile	3.7 KB	All	User profile management
SystemStatus	6.7 KB	Admin	Backend health monitoring

7. FEATURE COMPLETION STATUS

Feature Module	Backend	Frontend	Status	Notes
Authentication & RBAC	✓ Done	✓ Done	Complete	JWT + 6 roles enforced
Student Management	✓ Done	✓ Done	Complete	Full CRUD + admissions
Staff Management	✓ Done	✓ Done	Complete	—
Academic Structure (CBC)	✓ Done	✓ Done	Complete	PP1–Grade 9
Timetable	✓ Done	✓ Done	Complete	Visual weekly grid
Course Outlines	✓ Done	Partial	In Progress	Backend done; UI in academic pages
Course Resources	✓ Done	Partial	In Progress	File upload backend ready
Attendance (Daily)	✓ Done	✓ Done	Complete	4 statuses
Attendance (QR Code)	✓ Done	Partial	In Progress	Frontend UI incomplete
Assessments & Grading	✓ Done	✓ Done	Complete	CBC + traditional grades
Report Cards	✓ Done	✓ Done	Complete	PDF export available
Fee Management	✓ Done	✓ Done	Complete	Invoicing + payments

M-Pesa STK Push	✔ Done	Partial	In Progress	Backend service ready; not tested E2E
Announcements	✔ Done	✔ Done	Complete	Role-targeted
Direct Messaging	✔ Done	Partial	In Progress	Basic UI
AI Chat Assistant	✔ Done	Partial	In Progress	Needs API key config
Reminders	✔ Done	Partial	In Progress	Linked to Google/MS
Google Calendar	✔ Done	Partial	In Progress	OAuth wired
Microsoft Tasks	✔ Done	Partial	In Progress	MS Graph API wired
Audit Logging	✔ Done	N/A	Complete	Backend-only
Dashboard Analytics	✔ Done	✔ Done	Complete	KPIs, charts
Flutter Parent App	Not Started	Not Started	Pending	Future milestone
SMS Notifications	Not Started	Not Started	Pending	Future milestone
Automated Test Suite	Not Started	Not Started	Pending	Vitest configured

8. PROGRESS SUMMARY BY PHASE

Phase 1 — Foundation (COMPLETE)

- Database schema designed (24 models, 6 enums)
- Express app configured with full middleware stack
- JWT authentication with refresh token rotation
- Role-based access control on all 6 roles
- Student management module (CRUD + admissions workflow)
- Staff management module
- Academic structure (grades, classes, subjects)

Phase 2 — Core Academic (COMPLETE)

- Timetable management with weekly grid view
- Attendance tracking (daily marking)
- Assessment and CBC grading engine
- CBC competency scoring (EE/ME/AE/BE)
- Report card generation with rankings
- Course outlines (JSON-based weekly plan)
- Course resources (file upload)

Phase 3 — Finance & Communication (COMPLETE)

- Fee structures per grade per term
- Invoice generation with unique numbering
- Payment recording (Cash, M-Pesa, Bank Transfer)
- M-Pesa STK Push service (backend ready)
- Announcements with role targeting
- Direct messaging between users

Phase 4 — Advanced Features (IN PROGRESS)

- AI chat assistant backend  | Frontend UI partial
- QR code attendance backend  | Frontend UI incomplete
- Google Calendar OAuth integration  | Frontend partial

- Microsoft Tasks integration  | Frontend partial
- Cloud file storage for resources  (needs Cloudflare R2 or S3)

Phase 5 — Testing & Deployment (IN PROGRESS)

- Docker Compose for local development 
- Render deployment config 
- Cloudflare Pages config 
- Automated test suite  (Vitest configured, no tests written)
- CI/CD pipeline  (GitHub Actions not set up)
- Production smoke testing  (pending)

Phase 6 — Mobile & Expansion (NOT STARTED)

- Flutter parent mobile application
- SMS/Email notification service
- Multi-school tenancy support
- Advanced analytics and reporting

9. API ENDPOINTS SUMMARY

Authentication — /api/auth

Method	Endpoint	Description
POST	/register	Register new user account
POST	/login	Login with email/password, receive JWT tokens
POST	/refresh	Refresh access token using refresh token
GET	/profile	Get authenticated user's profile
PATCH	/profile	Update profile information
POST	/change-password	Update account password

Students — /api/students

Method	Endpoint	Description
GET	/	List all students with pagination and filters
POST	/	Create student profile and user account
GET	/:id	Get full student details
PUT	/:id	Update student information
DELETE	/:id	Remove student record
POST	/:id/approve	Approve student admission
POST	/:id/reject	Reject student admission
POST	/:id/promote	Promote student to next grade

Academic — /api/academic

Method	Endpoint	Description
GET	/years	List academic years
POST	/years	Create academic year
GET	/terms	List terms
GET	/grades	List CBC grades (PP1–Grade 9)
GET	/classes	List classes
POST	/classes	Create a class
GET	/subjects	List subjects
POST	/timetable	Create timetable slot
GET	/timetable	Get timetable slots

Fees — /api/fees

Method	Endpoint	Description
GET	/structures	List fee structures
POST	/structures	Create fee structure
POST	/invoices	Generate student invoice
GET	/invoices/:studentId	Get student invoices
POST	/payments	Record a payment
POST	/mpesa/initiate	Initiate M-Pesa STK Push
POST	/mpesa/callback	Handle M-Pesa payment callback

10. DEPLOYMENT CONFIGURATION

10.1 Backend — Render.com

Setting	Value
Service Name	matundu-sms-backend
Runtime	Node.js 20
Region	Oregon (us-west)
Build Command	npm install && npx prisma generate
Start Command	npx prisma migrate deploy && npm start
Database	PostgreSQL (matundu-db, free tier)
Cache	Redis (matundu-redis, free tier)
CORS Origin	https://matundu-frontend.pages.dev
Auth Rate Limit	15 requests max

10.2 Frontend — Cloudflare Pages

Setting	Value
Config File	frontend/wrangler.toml
Build Command	npm run build
Output Directory	dist
Target URL	https://matundu-frontend.pages.dev

10.3 Local Development — Docker Compose

Run 'docker-compose up -d' to start all services:

Service	Port	Description
PostgreSQL	5432	Database server
Backend API	3000	Express.js API server
Frontend	5173	Vite dev server

11. CHALLENGES AND SOLUTIONS

#	Challenge Encountered	Solution Applied
1	Supabase connection failures in production	Configured Prisma connection pooling and retry logic
2	Auto-seed running on every login causing data loss	Moved seeding to one-time setup; removed from login flow
3	Assessment scores not grouped by term in frontend	Refactored API query to include term relations; grouped in UI
4	Dashboard showing hardcoded placeholder values	Replaced all static values with live API calls
5	Redis rate limiting not working in free-tier Render	Added graceful fallback to memory-based rate limiting
6	QR code attendance UI incomplete	Backend service complete; frontend UI deferred to Phase 4
7	M-Pesa callback URL not accessible in local dev	Using ngrok for local callback testing; prod URL configured
8	Google/Microsoft OAuth redirect URIs mismatch	Documented required URIs for production deployment config

12. KNOWN ISSUES AND BLOCKERS

Priority	Issue	Area	Status
HIGH	M-Pesa STK Push not tested end-to-end in production	Fees	Pending
HIGH	AI chat service requires external API key configuration	AI Module	Pending
HIGH	No automated test suite written (Vitest configured only)	Testing	Not Started
HIGH	No CI/CD pipeline configured in GitHub Actions	DevOps	Not Started
MEDIUM	Google/Microsoft OAuth redirect URIs need production update	Integrations	Pending
MEDIUM	QR Code attendance frontend UI is incomplete	Attendance	In Progress
MEDIUM	Course Resources need cloud storage (Cloudflare R2 or S3)	Academic	Pending
MEDIUM	Redis rate limiting needs validation in production environment	Security	Pending

LOW	Timetable.jsx_snippet file is a leftover debug artifact	Frontend	Cleanup Needed
LOW	PDF report card layout needs design polish	Reports	Pending

13. NEXT STEPS

Immediate (Week 1-2)

- Write automated unit tests using Vitest for all service modules
- Complete QR code attendance frontend scanning UI
- Set up GitHub Actions CI/CD workflow for auto-deploy on push
- Configure production OAuth redirect URIs for Google and Microsoft
- Polish AI chat student-facing interface

Short Term (Week 3-4)

- End-to-end M-Pesa STK Push sandbox testing
- Integrate Cloudflare R2 for course resource file storage
- Complete Swagger/OpenAPI documentation for all endpoints
- Full production smoke test on Render + Cloudflare Pages

Long Term

- Develop Flutter parent mobile application
- Integrate Africa's Talking or similar SMS gateway for notifications
- Database query optimization and API performance profiling
- Security audit and penetration testing
- Extend architecture for multi-school tenancy support

14. REFERENCES

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